Set → collection of elements.

$A = \{1, 2, 3\}$ $B = \{x, y, z\}$

$A \times B$ → cartesian product of A and B

$A \times B = \{(1,x), (1,y), (1,z), (2,x), (2,y), (2,z)\}$

Binary relation → where the first (a) element belongs to A and second belongs to (b) B

$R = \{(1,x), (a,y)\}$

$R = \{(1,x), (1,y)\}$

(a, b)

is just a way to show a connection b/w two sets.

$A = \{0, 1, 2\}$ $B = \{a, b\}$

$A \times B = \{(0,a), (0,b), (1,a), (1,b), (2,a), (2,b)\}$

→ everything we can possibly include in a relation.

(P)

A Relation can include

1) None – empty relation
2) Some –
3) All – full cartesian product

Relation on a set is just $A \times A$.

$A = \{1, 2, 3\}$

$(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)$

Relation on $A = \{(1,1), (2,1), (2,2)\}$



Note: a divides b → $b \div a$

e.g How many relations are there on a set with n elements ?

6

$A = \{a, b, c\}$

$A \times A = \{(a,a), (a,b), (a,c), (b,a), (b,b), (b,c), (c,a), (c,b), (c,c)\}$

A^2

→ $|3|^2$

→ (9) pairs

(Properties of Relation)

Reflexive

→ A relation is reflective when every element is related to itself meaning a and b are same

$A = \{1, 2, 3\}$

$A \times A = \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)\}$

these pairs are reflective.

Reflective, each element is related to itself

For a Relation set to be reflective it must contain all pairs of (a,b) where a is similar to b while also containing a is not similar to b

Symmetric relation: is when both (a,b) and (b,a) is in R

if a is related to b then b must also be related to a.

Both (a,b) and (b,a) must be in the set

$A \times A = (1,1)\ (1,2)\ (1,3),$
$(2,1), (2,2), (2,3),$
$(3,1), (3,2), (3,3)$

$R = \{(1,1), (1,2), (2,1), (2,2), (3,3)\}$

$(1,2) \longleftrightarrow (2,1)$
$(1,1) \longleftrightarrow (1,1)$

(3,3) is also Symmetric

Not Symmetric $R = \{(1,2), (2,3), (3,3)\}$

Antisymmetric Relation: A relation is said to be Antisymmetric if it contains all (a,a) some other (a,b) but not (b,a) together like in Symmetric.

$R = \{(1,1), (2,2), (1,2), (2,1)\}$ (X)

$R = \{(1,1), (2,2), (1,2)\}$ (✓)

A relation R is transitive if a is related to b and b is related to c then a must also be related to c.

$A \times A = \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)\}$

Transitive Relation

$R = \{(1,2), (2,3), (1,3)\}$

if 1 is related to 2 and 2 is related to 3 the 1 is also related to 3.

$R = \{(2,1), (3,1), (3,2), (4,1), (4,2), (4,3)\}$

$(3,2) \rightarrow (2,1) \rightarrow (3,1)$ —

$(4,2) \rightarrow (2,1) \rightarrow (4,1)$ —

$(4,3) \rightarrow (3,1) \rightarrow (4,1)$ —

if all pairs are transitive then the Relation is said to be transitive.

Combining Relations:

$R = \{(3,2), (2,1), (3,1), (4,1)\}$

R is not transitive

Two Relations from Set A to B can be combined in any way as Set A and Set B can be combined.

Union → combines everything | Intersection → common |

Set A = {1, 2, 3} Set B = {1, 2, 3, 4}

$R_1 = \{(1,1), (2,2), (3,3)\}$
$R_2 = \{(1,1), (1,2), (1,3), (1,4)\}$

$R_1 \cup R_2 = \{(1,1), (1,2), (2,2), (1,3), (3,3), (1,4)\}$

$R_1 \cap R_2 = \{(1,1)\}$

$R_1 - R_2 = \{(2,2), (3,3)\}$ what is in R_1 but not in R_2

$R_2 - R_1 = \{(1,2), (1,3), (1,4)\}$

OR ($\cup$) → one or both can be true
XOR ($\oplus$) → exactly one can be true but not both

$R_1 = \{a, b\}$ $R_2 = \{b, c\}$

$R_1 \cup R_2 = \{a, b, c\}$
$R_1 \oplus R_2 = \{a, c\}$ → (exclude items that are in both)

Another method to find $\oplus$ = union − intersection

A relation that is reflexive, symmetric, transitive is an equivalence relation.

Composite Relation: ???

R is a relation from A to B → (a, b)

S is a relation from B to C → (b, c)

S∘R union is a relation from A to C (a, c)

$A = \{1, 2\}$ $B = \{x, y\}$

$C = \{P, Q\}$

$R = \{(1, x), (2, y)\}$ $S = \{(x, P), (y, Q)\}$

Representing Relations:

$R = \{(a_i, b_j)\}$ — column ↑, row ↓

if a pair of (a_i, b_j) is in R write 1 and if it is not write 0.

$A = \{1, 2, 3\}$ $B = \{1\}$

R contains (a, b) whenever a > b

$A \times B = \{(1,1), (1,2), (2,1), (2,2), (3,1), (3,2)\}$

r x c

$R = \{(2,1), (3,1), (3,2)\}$ (i, j)

$$\text{Matrix} = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cc} 1 & 2 \end{array} \\ \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} \end{array}$$

3 x 2

A = {a1, a2, a3} B = {b1, b2, b3, b4, b5}

$$M_R = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \end{array} \\ \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{pmatrix} \end{array} \quad (3 \times 5)$$

listing all the pairs in the Relation

(a_2, b_1), (a_1, b_2), (a_3, b_1), (a_2, b_3)

(a_3, b_3), (a_2, b_4),

(a_3, b_5)

→ Not Reflective

→ not Symmetric

→ Anti-Symmetric

In Matrix

Reflective relation → all diagonal entries are 1.

Symmetric relation → $M_R = M_R^T$

Asymmetric relation →



look above and Below the main diagonal

→ if there is mirrored zeros and no mirrored 1 then → Anti Symmetric

→ if there is all 1's mirrored then NOT Anti-Symmetric

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$M_{21} \neq M_{12}$

two diff elements related both ways.

Hence this is not anticymmetric

Union → OR

Intersect → AND

$$M_{R1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \qquad M_{R2} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$R_1 \cup R_2 = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$R_1 \cap R_2 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$a_0 = 2$, $a_1 = 7$

$a_n = a_{n-1} + 2a_{n-2}$

$a_n = r^n$

$r^n = r^{n-1} + 6r^{n-2}$

divide by r^{n-2}

$\frac{r^n}{r^{n-2}} = \frac{r^{n-1} + 6r^{n-2}}{r^{n-2}}$

$r^2 = 5ar + b$

$r_1 = 2 \quad r_2 = 3$

$r^2 - 5r + 6 = 0$